

ProCPX – Functional Requirement Document

Phase 1 – Procurement, BOQ, Commercial Evaluation, AI Cost Intelligence & Work Order / PO

Revision: 1.0 | Functional baseline based on the revised process flow

1. Objective

ProCPX shall provide an integrated procurement platform that starts with Purchase Request and manages BOQ creation, PR approval and acceptance, commercial price checking, vendor quotation comparison, AI-driven market/benchmark pricing, AI cost analysis and cost break-up, vendor negotiation, final comparison, PPO creation and approval, and issuance of the Work Order / PO.

The revised Phase-1 flow deliberately excludes project budget upload, budgetary controls, budget consumption/shortfall workflows, vendor bill processing, PO/Non-PO billing controls, execution, measurement/certification and project-closure processes.

2. Scope

- Purchase Request and cost-centre/category mapping.
- BOQ creation through GFC(Good For Construction) → BOQ, Standard BOQ or New BOQ Upload.
- PR approval and PR acceptance.
- Commercial price check using Rate Card Price, Vendor Quote Comparison and Market / Benchmark Price.
- Price comparison and price-acceptable decision.
- AI-supported Cost Analysis and Cost Break-up when pricing is not acceptable.
- Vendor negotiation, counter-offer/counter-quotation tracking and negotiation history.
- Final comparison, PPO creation and PPO approval.
- Work Order / PO generation and issuance.
- Role-based access, audit trail, notifications, dashboards and reporting.
- AI data capture and intelligence framework for BOQ, market pricing and cost analysis.

3. Revised End-to-End Process

Step	Process	Functional Purpose
1	Purchase Request	Starting point of the workflow.
2	Cost Centre / Category Mapping	Associate PR with the relevant cost centre and procurement category; no budget validation.
3	BOQ Creation	GFC → BOQ (AI-assisted), Standard BOQ or New BOQ Upload.
4	PR Approval	Approval of requirement and BOQ.
5	PR Acceptance	Buyer/procurement acceptance for commercial processing.
6	Commercial Price Check	Rate Card Price + Vendor Quote Comparison + Market / Benchmark Price.
7	Price Comparison	Compare all applicable commercial reference prices.
8	Price Acceptable?	Yes → Final Comparison. No → AI Cost Analysis → Cost Break-up → Manual/Vendor Negotiation.
9	Vendor Negotiation	Record negotiation, counter-offers and revised prices.
10	Final Comparison	Compare final negotiated offers and recommend vendor.
11	PPO Creation	Create Purchase Price Offer / internal commercial approval document.
12	PPO Approval	Approve or reject PPO.
13	Work Order / PO Issued	Generate and issue Work Order / Purchase Order to selected vendor.

4. Functional Modules & Requirements

ID	Module	Purpose	Key Functional Requirements	Primary Role
FR-01	Cost Centre & Master Setup	Maintain cost centres, categories, services, item groups, units and user mappings.	Create/edit cost centres; map categories; assign users/roles; activate/deactivate masters.	Buyer Admin
FR-02	Purchase Request	Create the procurement requirement.	PR number; requester; project/reference; cost centre; category; required date; item/service details; specifications; quantity; attachments; remarks; submit.	Requester / Buyer
FR-03	BOQ Creation	Create BOQ using three supported routes.	GFC → BOQ; Standard BOQ selection; New BOQ Upload; item validation; quantity/unit/specification mapping; revision control.	Estimator / Buyer
FR-04	PR Approval	Approve the PR and BOQ before commercial processing.	Approval matrix; approve/reject; comments; resubmission; status history.	Approver
FR-05	PR Acceptance	Accept approved PR for procurement action.	Review PR; accept/reject; assign buyer; start commercial evaluation.	Buyer
FR-06	Commercial Price Check	Establish commercial reference prices.	Rate Card Price; Vendor Quote Comparison; Market / Benchmark Price; internal standard price where available.	Buyer
FR-07	Price Comparison	Provide a normalized comparison of offers and benchmarks.	Compare item-wise and total prices; variance; lowest price; percentage difference; historical/reference comparison.	Buyer
FR-08	AI Cost Analysis	Analyze prices where the price is not acceptable.	Material, labour, equipment and OH modelling; rate reasonableness; historical intelligence; variance analysis.	Buyer
FR-09	AI Cost Break-up	Explain the expected cost composition.	Cost component breakdown; benchmark range; quantity/rate impact; assumptions; AI confidence/exception flag.	Buyer
FR-10	Vendor Negotiation	Manage negotiation with vendors.	Negotiation event log; original quote; counter-offer; revised quote; reason; target price; buyer remarks; vendor response; attachments.	Buyer
FR-11	Final Comparison	Finalize commercial recommendation.	Compare negotiated offers; commercial ranking; recommendation; approval comments; final selected price.	Buyer / Approver
FR-12	PPO Creation	Create the internal Purchase Price Offer.	Pull approved PR/BOQ/vendor data; negotiated price; terms; taxes; delivery; payment;	Buyer

			validity; attachments.	
FR-13	PPO Approval	Approve PPO before order issuance.	Approval workflow; approve/reject; comments; version control; audit trail.	Approver
FR-14	Work Order / PO Issuance	Issue the final order.	Generate Work Order / PO number; vendor details; BOQ/line items; negotiated rates; terms; delivery; approval references; PDF generation; vendor communication.	Buyer

5. AI Functional Requirements

ID	Capability	Requirement
AI-01	GFC → BOQ	Extract quantities from GFC drawings/documents, map specifications and units, identify missing/duplicate items, and propose a structured BOQ for user validation.
AI-02	BOQ Intelligence	Map extracted items to ProCPX standard items, rate-card references and category taxonomy; flag low-confidence mappings.
AI-03	Market / Benchmark Price	Provide market-aligned benchmark pricing using available buyer/seller data, historical transactions, market inputs, regional factors and trend information.
AI-04	Price Reasonableness	Calculate variance against rate card, vendor quotation, benchmark and available historical/internal standard prices.
AI-05	Cost Analysis	Model cost components such as material, labour, equipment and overheads for items requiring negotiation support.
AI-06	Cost Break-up	Show estimated component-wise cost and explain the key drivers of quoted price.
AI-07	Negotiation Intelligence	Suggest negotiation reference/target ranges and highlight areas where quoted price materially differs from expected cost.
AI-08	Human-in-the-loop	AI outputs must be reviewable, editable and accepted/rejected by an authorized user; AI must not silently alter approved commercial values.

6. Commercial Evaluation Rules

- Every comparable item should show applicable Rate Card Price, Vendor Quote Price and Market / Benchmark Price.
- Where an internal standard price exists, it shall be displayed as an additional comparison reference.
- The system shall calculate absolute and percentage variance against each applicable reference.
- If the price is acceptable, the workflow proceeds to Final Comparison.
- If the price is not acceptable, the buyer shall be able to invoke AI Cost Analysis and Cost Break-up.
- AI analysis shall support negotiation and shall not automatically finalize a vendor.
- Vendor Negotiation shall retain original quotation, each counter-offer, revised quotation, date/time, user and comments.
- Final comparison shall use the latest valid negotiated commercial values.
- PPO shall be created only after the commercial recommendation is finalized.
- Work Order / PO shall be issued only after PPO approval.

7. User Roles & Access

Role	Access / Responsibility
Requester	Create and submit PR; view own PR status; respond to clarification.
Buyer / Procurement	Manage BOQ, commercial comparison, vendor quotations, AI analysis, negotiation, final comparison and PPO.
Estimator / BOQ User	Prepare/review BOQ and validate AI-generated BOQ.
Approver	Approve/reject PR and PPO according to configured approval matrix.
Vendor	Submit quotation and negotiation responses through the permitted vendor interface/portal.
Buyer Admin	Manage masters, users, roles, approval matrices and configuration.
System / AI Service	Process documents/data and generate recommendations; no final approval authority.

8. Key Data Objects

- Purchase Request: PR number, requester, cost centre, category, requirement, specification, quantity, attachments, dates and status.
- BOQ: item code, description, specification, unit, quantity, source, revision, AI confidence and validation status.
- Vendor Quote: vendor, quotation number/date, item-wise rate, taxes, lead time, terms, validity and attachments.
- Rate Card: item/category, standard rate, unit, effective date, geography and revision.
- Benchmark: benchmark rate/range, source, geography, date, confidence and methodology.
- Cost Analysis: material, labour, equipment, overhead, assumptions, benchmark and AI recommendation.
- Negotiation: event number, vendor, original rate, counter-offer, revised rate, target, comments, timestamp and user.
- PPO: PR/BOQ reference, selected vendor, negotiated values, terms, approvals and version.
- Work Order / PO: order number, vendor, final BOQ/rates, commercial terms, delivery requirements and issue date.

9. Notifications & Workflow

- PR submitted → approver notification.
- PR approved → buyer notification.
- PR rejected → requester/buyer notification with comments.
- Quotation submitted/updated → buyer notification.
- Negotiation counter-offer → relevant buyer/vendor notification.
- PPO submitted → approver notification.
- PPO approved → buyer/vendor notification and Work Order / PO generation.
- All notifications shall be available through in-app notification; email/SMS/other channels should be integration-ready.

10. Audit Trail & Controls

- Every material change shall capture user, date/time, old value, new value and reason where applicable.
- Quotation and negotiation revisions shall be immutable once submitted, with a new revision created for changes.
- AI recommendations shall retain the input snapshot, output, confidence/exception information and user action.
- Approval actions shall capture approver, timestamp, decision and comments.
- Documents shall retain version history and access permissions.

11. Dashboards & Reports

- PR status and ageing dashboard.
- BOQ creation and AI validation dashboard.
- Rate Card vs Vendor Quote vs Benchmark variance report.
- Negotiation savings / variance report.
- Vendor-wise quotation and negotiation performance.

- Category-wise price intelligence.
- AI recommendation acceptance/override report.
- PPO and Work Order / PO status report.
- Full procurement cycle audit report.

12. Non-Functional Requirements

- Responsive web application with role-based access control.
- Secure authentication and authorization.
- Encryption of data in transit and at rest.
- Scalable architecture for growing buyer/seller datasets and AI processing.
- High availability and automated backup/recovery.
- Search, filtering, sorting, export and document download.
- API-first integration capability for ERP/finance, vendor portals, e-signature and notification services.
- Performance target: normal transactional screens should respond within an agreed SLA; large AI/document processing may be asynchronous with status tracking.

13. Integration Requirements – Future Ready

- ERP / Finance system for vendor, item, PO and financial master synchronization.
- Vendor portal for RFQ/quotation and negotiation exchange.
- Document storage / DMS.
- E-signature service.
- Email, SMS and in-app notification services.
- AI/ML services for document extraction, benchmark intelligence and cost modelling.
- Analytics / BI platform.

14. Acceptance Criteria – Phase 1

- A user can create and submit a Purchase Request without any budget upload or budget check.
- Approved PR can be converted into a valid BOQ using GFC → BOQ, Standard BOQ or New BOQ Upload.
- PR approval and acceptance are fully auditable.
- Commercial comparison displays rate card, vendor quote and benchmark references where available.
- The system clearly identifies acceptable vs non-acceptable pricing.
- AI Cost Analysis and Cost Break-up can be invoked for non-acceptable pricing and the output can be reviewed/overridden.
- Vendor negotiation history is retained with all quotation revisions.
- Final comparison reflects the latest valid commercial data.
- PPO can be created and routed for approval.
- After PPO approval, the system generates and issues a Work Order / PO.
- No budgetary control, budget upload, budget consumption or post-order billing workflow is triggered in this Phase-1 flow.
- Role-based access, notifications and audit trails work across the complete process.

15. Explicit Out-of-Scope for This Phase

- Project Budget Upload.
- Cost Centre-wise Budget.
- Budget Availability / Budget Shortfall.
- Budget Consumption / Fresh Budget Request.
- Final Budget vs Actual.
- Vendor Bill Submission.
- PO / Non-PO Bill Check.
- Final Bill Value and Final Bill Approval.
- Execution, Measurement / Certification and Project Closure.

16. Phase-1 Development Priority

- Priority 1 – PR, BOQ, approval/acceptance and commercial comparison foundation.
- Priority 2 – Rate Card, vendor quotation comparison and benchmark pricing.
- Priority 3 – AI GFC → BOQ, market intelligence, cost analysis and cost break-up.
- Priority 4 – Negotiation, final comparison, PPO workflow and Work Order / PO issuance.
- Priority 5 – dashboards, reports, integrations, audit and administration.

17. Final Functional Flow

**PURCHASE REQUEST → BOQ (GFC / STANDARD / NEW UPLOAD) → PR APPROVAL → PR ACCEPTANCE →
COMMERCIAL PRICE CHECK → PRICE COMPARISON → PRICE ACCEPTABLE? → FINAL COMPARISON / AI COST
ANALYSIS & COST BREAK-UP → VENDOR NEGOTIATION → PPO CREATION → PPO APPROVAL → WORK ORDER /
PO ISSUED**